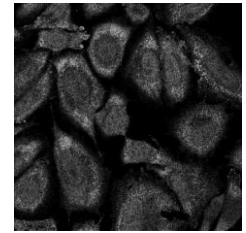
(a) Knowledge-level Biological Context

What is cytosol?

What are SiHa cells?

Cytosol is the aqueous component of the cytoplasm outside membrane-bound organelles and…
SiHa cells originate from human cervical tissue, belong to the squamous epithelial lineage…

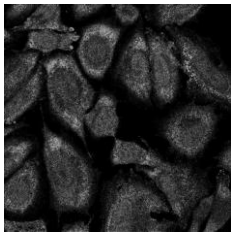
(b) Detail-level Visual Description



Please describe the visual characteristics.

A **grayscale** fluorescence microscopy image showing **densely packed**, **irregularly shaped cells**. The intracellular signal is **diffuse and finely granular**, with some **brighter central regions** and **darker internal spots**. Cell boundaries are fairly distinct…

(c) Category-level Biological Identification



Identify the subcellular structure and cell line with visual evidence.

GT: Cytosol of SiHa

The image most likely shows **nucleoplasmic/nuclear** localization in **U-2 OS cells**.
•Subcellular structure: nucleoplasm (nucleus). The fluorescence occupies large oval-to-irregular…
•Cell line: U-2 OS, tentatively. The nuclei are large, flattened…

(d) Recognition Across Distribution Shifts

